

## Digital Astronomy Eyepiece App

Version 0.700, 21.09.2026

Copyright Andriy Melnykov 2026

This version supports ZWO ASI, SVBony, ToupTek cameras

Libraries used (see also licenses folder):

- ASICamera2 SDK, copyright ZWO company
- SVBCamera SDK, copyright SVBony company
- ToupTek SDK, copyright ToupTek Astro
- OpenCV, Apache License 2.0
- CFITSIO, copyright National Aeronautics and Space Administration
- frugally-deep, copyright (c) 2016 Tobias Hermann
- OpenCV reg module, copyright (C) 2013, Alfonso Sanchez-Beato

This app is developed for fast and ultra-fast EAA observations (Electronic Assisted Astronomy), mainly to be used with very fast optical systems in range F1.0-F3.0. It meant to be somewhere in between EAA and night vision, providing more real-time feeling and using exposure times in range 200ms .. 4s and integration time under 1 minute. The app can be of course used also with slower optics and longer exposures, and also with night vision devices, equipped with cameras.

## Installation

No installation needed, simply copy the release folder.

**Check the \*.exe and \*.dll files after download with antivirus software!**

## Folder content

Digital\_Eyepiece\_APP.exe – the app itself, start this file.

ASICamera2.dll, SVBCameraSDK.dll, toupcam.dll, opencv\_world480.dll, cfitsio.dll, zlib.dll – libraries needed for the app.

config.txt – configuration file with parameters.

533\_dark\_400ms\_g500\_off100\_25C.fits, 533\_dark\_4s\_g300\_off100\_25C.fits, MasterFlat\_hyperstar\_bin2.fits – examples of calibration files, darks for “video” and “foto” mode, flat for both modes (filenames are also examples).

Readme.pdf – this file with description and instructions.

saved\_pictures – folder, where images will be saved during observation. If not exists, will be created.

licenses – folder with licenses.

log.txt – debug output file. If not exists, will be created, if log/debug option is activated.

train\_xx.json, train\_xx.onnx – neural network configurations for AI noise reduction.

## Controls

The key idea of control is simplicity. Control of the app during execution is done via keyboard or simple GUI. Control keys can also be changed in configuration file. Alternatively there are predefined hotkeys that can be used by external controller.

For key control to be active focus must be on the image window. Hotkeys are active without focus.

“x” – for “eXit”, close the app

“m” or Ctrl+Alt+F16 – for “Mode”, switches between fast, “video” mode, and stacking, “foto” mode

“+” or Ctrl+Alt+F14, “-” or Ctrl+Alt+F13 – control of stretching and brightness, similar to gain of night vision device

“z” or Ctrl+Alt+F17 – for “Focusing”, image zoom for focusing in “video” mode, image zoom in “foto” mode

“p” or Ctrl+Alt+F15 – for “Palette”, switching between normal RGB and alternative (for example dual-band) palette

“s” – for “Save”, save images

“b” or Ctrl+Alt+F18 – for “Background”, enables or disables additional background correction

GUI with on-screen buttons can be activated or deactivated per options.

Control keys and GUI are sometimes not very responsive, many calculations are done after pressing buttons. The SW is still usable, just don't rush. Clock waiting symbol can be activated/deactivated per options to be informed, if software is still processing the image.

### Modes, key “m”

There are two modes, first “video” mode meant for searching for the object, constellation, stars, also for focusing, etc. Typically, short exposure, high gain, simple noise reduction can be activated. Second “foto” mode (from German Foto=photo) is for stacking. Typically, but not necessary, longer exposure, lower gain, more sophisticated noise reduction.

Stretching, keys “+”, “-”

Control of non-linear stretching, works also similar to gain of night vision devices, but without changing the “gain” parameter of the camera.

Focusing/Zoom, key “z”

Activates zoom of the image with configurable appearance (RGB or center/edges), for more convenient focusing in “video” mode.

Activates zoom of the image in “foto” mode.

Switching palettes, key “p”

Switches between configurable color palettes including white balance and color correction matrix, for example for different filters. This way, for example for observation with dual-band filter, the red of H-Alfa can be shifted to orange-gold, cyan of O-III to deep blue, enhancing the colour contrast dramatically, and making H-Alfa objects better visible. There are also other uses of this function, i.e. black/white from colour, H-Alfa to black/white, imitation of greenish or blueish screen of night vision devices, etc.

Save images, key “s”

Saves three images – one as seen on display, second with processing, but with full resolution, third only stacked image without processing, and one text file with used parameters. Images are saved in the folder “saved\_pictures”, which is automatically created, if not exists.

Background correction, key “b”

Enables or disables additional background correction if the background appears blotchy, especially near the edge. This function works correctly only when the object is smaller than the field of view.

## **Important features**

One of the most important features of the app is automatic background subtraction, with two variants: constant level and gradient. Black level subtraction and stretching is based on iterative approach, described by Roger N. Clark, [clarkvision.com](http://clarkvision.com). Black level is determined (for each of RGB colours separately) as a point with relatively low threshold of histogram from left side, the value is configurable. Automatic background subtraction allows high offset parameter in the camera to not lose dark details and also wideband observations under light polluted sky. For background subtraction precise flat frame is essential, also precise dark frame is important to apply flat frame properly. Black point can be additionally shifted to taste with black point offset parameter.

For systems with image circle smaller, than the sensor size (i.e. “Ultra-fast EAA”) and for systems with strong vignetting a circular mask can be defined to limit the region for black level calculation and image alignment. This circular mask can be temporarily shown for adjusting purpose.

Image can also be zoomed to display small image circle in full image height.

AI noise reduction can be very effective for EAA. The app contains experimental version of AI noise reduction filter. It is trained on my own images – can be less effective on others.

The app can output images on a second display, used as electronic eyepiece. Different types of electronic eyepieces can be configured, for bino or mono mode.

The app can be also used in eyepiece-only mode.

## **Other features**

Banding filter can be activated, if brightness bands are visible, which sometimes happens at short exposures.

Image rotation, resize to display size, zoom can be used to adjust displayed picture.

Bilateral noise reduction filter can be activated, separately for video and foto mode.

Color correction matrix is available besides simple white balance – can be very helpful to reach natural color rendition of the sky, but needs special calibration with color chart.

Highlight protection feature is available – sort of HDR tonemapping for highlights, prevents saturation of bright objects by stretching.

There are a set of image enhancement functions: midtone contrast, sharpen.

The app can be called from command line with a parameter, containing filename for config file. This way many different configurations can be switched fast. If called without parameters, default config file is “config.txt”.

Filenames of calibration files can be changed in config file. Combined with config filename in command line parameter, many different configurations can be switched fast.

Black screen can be activated with mouse click on image in main window.

### **Some other interesting features**

Optical mask – black circular mask, makes picture more optic-device-like. Can also be used with small image circle to hide the white background after flat frame correction and reduce visible image to image circle. If activated, image is always cropped to square.

Enhancing bright stars – can make bright stars more visible and colorful, somewhat similar to starglow filter, or halo by night vision devices.

### **Experimental Night Vision mode**

This mode allows fast (e.g. 10fps) video with different noise reduction strategies. Contains also motion detection, differentiation of noise reduction depending on motion, ultra-fast neural network for noise reduction. Works with monochrome cameras. Not recommended and not tested with color cameras.

### **Further ideas**

One can make an Arduino-based or other simple DIY keyboard, USB or Bluetooth®, with only the keys needed to control the app (or hotkeys), and use it as a hand controller.

### **Configuration file**

Contains parameter, that can be adjusted to adapt the app to your system.

If you need configurations for different systems, simply copy the whole app folder or use command line parameter as config filename.

**any //camera name should contain this string, or "any" for first found camera**

Here a part of a camera name can be defined, to access a specific camera, if many are connected to the system. For example "ASI533" for ASI533mc camera. Exact camera names can be checked in log files / debug output. If "any" – first found camera will be used.

**0 // debug output, screen and log file: 0 - none, 1 - active**

Debug output into log file. Activate, if problems occurring, log file can be analyzed or sent for analysis. Performance is worse, if this option is activated, don't use, if not needed.

**0 15 // save pictures automatically yes/no: 1/0, after n-th frame**

Pictures can be saved automatically – useful feature for example for public events, to not forget to save pictures in a rush. First parameter is for activation, second is the interval as number of stacked frames.

**//----- Camera and preprocessing Parameters for video mode**

**400 //exposure time for video mode, ms**

**500 //gain for video mode**

**50 //White balance Red for video mode**

**50 //White balance Green for video mode //not all cameras use it**

**50 //White balance Blue for video mode**

**100 //offset for video mode**

**0 //high speed mode for video mode, 0 - not active, 1 - active**

Usual parameters of a camera. I prefer to have high gain here, exposure as a balance of visibility of objects and reaction speed. Offset is always high to not loose dark details, histogram of single dark frame must be checked in other software, that no (few) black pixels are produced. White balance adjusted ideally to produce same peaks for all colours in dark frame.

**1 //use dark frame for hot pixel correction, for video mode, 0 - don't use, 1 - use**

**1 //subtract dark frame from light frame, for video mode, 0 - don't use, 1 - use**

Dark frame calibration can be activated with two variants: only interpolating of hot pixels, based on dark frame (only first parameter activated) or complete dark frame subtraction (both parameters activated).

**0 //use flat frame for video mode, 0 - don't use, 1 - use**

Flat frame correction can be activated here for video mode. Not recommended for noisy "video" mode.

**1 //banding filter, 0 - disabled, 1 - rows, 2 - columns**

**100 // banding filter window size, 3..200**

**1.2 // banding filter threshold, 1.0 .. 2.0**

Banding filter can be activated here. Threshold must be started with 1.0 and slowly increased till banding filter is effective. Too high values of threshold are causing dark bands on bright objects.

**//----- Camera and preprocessing Parameters for foto mode**

**4000 //exposure time for foto mode, ms**

**300 //gain for foto mode**

**50 //White balance Red for foto mode**

**50 //White balance Green for foto mode //not all cameras use it**

**50 //White balance Blue for foto mode**

**100 //offset for foto mode**

Usual parameters of the camera. Exposure depends on your system and preferences, gain must be adjusted to get comparable brightness between two modes, without changing the stretch. Offset is always high, histogram of single

dark frame must be checked in other software, that no (few) black pixels are produced. White balance adjusted ideally to produce same peaks for all colours in dark frame.

```
1 //use dark frame for hot pixel correction, for foto mode, 0 - don't use, 1 - use
```

```
0 //find hot pixels without/additional to dark frame
```

```
1 //subtract dark frame from light frame, for foto mode, 0 - don't use, 1 - use
```

Dark frame calibration can be activated with two variants: only interpolating of hot pixels, based on dark frame (only first parameter activated) or complete dark frame subtraction (first and third parameters activated).

Also, hot pixels can be found additionally without dark frame (second parameter) – experimental function, can produce artefacts. Try, if you see some hot pixels after applying dark frame.

```
1 //use flat frame for foto mode, 0 - don't use, 1 - use
```

Flat frame correction can be activated here for foto mode.

```
//----- Calibration files
```

```
533_dark_400ms_g500_off100_25C.fits //dark for video mode
```

```
533_dark_4s_g300_off100_25C.fits //dark for foto mode
```

```
MasterFlat_hyperstar_bin2.fits //flat for both video and foto mode
```

Filenames of calibration files can be changed here. Filenames are examples.

```
0 // use spline flat correction
```

```
0.0 0.7 1.0 //radius points for spline flat correction
```

```
1.000 1.000 1.000 //R channel gain value points for spline flat correction
```

```
1.000 1.000 1.000 //G channel gain value points for spline flat correction
```

```
1.000 1.000 1.000 //B channel gain value points for spline flat correction
```

Additional correction of flat frame, mostly for residual bright edges. Contains reference points in radial direction, and corresponding gains for each color. For special cases, leave deactivated otherwise.

```
//----- Parameters for both video and foto mode
```

```
0 //monobin for both video and foto mode - only 0 supported for now!
```

```
2 //bin for both video and foto mode, 1, 2, ..
```

Bin recommendation is to get image in range of 700-1500 pixels. For (Ultra-)fast EAA it is generally a good idea to sacrifice resolution for sensitivity. Performance of the app is better with smaller pictures.

2 //bytes per pixel, 1 for RAW8, 2 for RAW16 - only RAW16 supported for now!

90 //bandwidth

No need to change.

20 // sigma for dark frame hot pixel correction, 10 .. 50

Threshold for hot pixel detection in dark frame. Higher the threshold – less hot pixels detected.

0 // ROI zoom activation, 0/1: off/on

Using only half of the sensor (central part only) as image source. Darks and flats must be captured accordingly.

5000 0 // internal image for processing: scale down to height (pixels), crop to square (0/1 on/off)

Scaling down of internal image to defined height (no scaling up, if parameter > image height – no effect), and activating of square cropping internal image. Higher sensitivity can be achieved (like binning) and higher processing speed.

0.1 0.1 //blackpoint correction for monitor, x1, y1 of corner point (no correction for x1=y1)

0.1 0.1 //blackpoint correction for eyepiece, x1, y1 of corner point (no correction for x1=y1)

Black point correction of corresponding display. Linear interpolation between (0,0) and (x1,y1), and between (x1,y1) and (1,1). For example 0.1 0.05 makes near black (0.1 value) darker.

//----- Other parameters

10 //cooler target temperature, °C

0 //cooler activation, 0 for cooler off, 1 for cooler on

Self explanatory.

800 //display size, display window height in pixels

Vertical size of the image to be displayed on main screen.

0 //image flip horizontally: 0 - none, 1 - flip

2 //image rotation for display: 0, 1, 2, 3 (0, 90°, 180°, 270° clockwise)

Flip and rotate to adjust image orientation.

**2 //background subtraction: 0 - none, 1 - constant level, 2 - gradient**

**0.05 0.01 //normalised histogram level from left, for background subtraction, for video and foto mode**

Gradient method is recommended. 0.05 and 0.01 (separate parameters for “video” and “foto” modes) in the second line mean the very left part of histogram is assumed as black level – preferred for showing dark details.

**0.02 //blackpoint offset, added after background subtraction**

The offset, added after background subtraction – helps to reveal more faint details, if background is too black.

**1 //circular mask as area for background subtraction (for small image circle): 0 - none, 1 - active**

**0.9 //circular mask for background subtraction diameter, parts of image height**

**0 //show circular mask for background subtraction for test: 0 - none, 1 - active**

It is recommended to set parameters like this per default, to limit area of black level detection. You can change the size of the mask with the second parameter, and make the mask visible with the third parameter.

**1 //noise reduction, for both video and foto mode: 0 - none, 1 - bilateral filtering**

**0.3 0.0 //noise reduction strength for video and foto mode, 0..1**

Self explanatory. Zero in the second parameter also deactivates noise reduction.

**30 0.2 0.5 // midtone contrast enhancement for foto mode, detail radius, midtone width, strength, deactivated if strength=0**

**1.5 0.0 // sharpen enhancement, sigma (size), amount, deactivated if amount=0**

Experimental functions to enhance midtone contrast and sharpness.

**1 //circular mask for display (optic like): 0 - none, 1 - active for both video and foto mode**

Here the black circular mask can be activated on main screen, making picture more optic-device-like, or hiding not usable parts of small image circle.

**50.0 //initial stretching factor: 1.0 is neutral, higher value is more stretching**

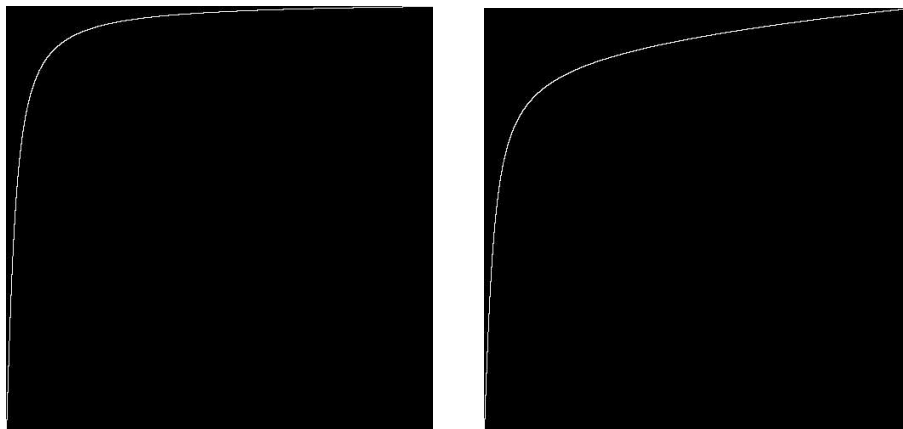
Initial stretching, can be adjusted to preference. 50.0 is already a strong stretching, recommended as default.



**0.9 0.9 //star protection factor for stretching, 1 - no effect, 0.5-0.99 - stars are more linear, nebulas darker (for nebula, for stars)**

Additional factor to control highlights part of the stretching curve. 1.0 has no additional effect (=deactivated), values <1.0 produce more linear curve in highlights range, to protect stars or bright nebulae parts from saturating and achieve more natural image.

Compare the curves with stretching factor 50.0 and star protection factor 1.0 (left) and 0.8 (right):



**Palette: UV/IR**

Begin of first color palette block. The name “UV/IR” is showing in status field, if status field is active.

**0.5 //mix between rgb stretch (0.0) and luminance chroma preserving stretch (1.0)**

Mix between two stretching algorithms – rgb stretch (simple nonlinear per-channel stretch, less color in highlights) and chrome preserving stretch (may be too much color in highlights). Defined for each palette separately.

**1.43 1.0 1.77 // White balance correction for UV/IR palette**

White balance correction for each palette. Can be adjusted to have natural or preferred colours of sky objects, or calibrated with color chart. Deactivated, if 1 1 1. Defined for each palette separately.

1.43 1.0 1.77 is an example for ASI533MC with Optolong UV/IR filter.

**0 // use color correction matrix for RGB palette, 0 - don't use, 1 - use**

**1.39 -0.36 -0.03 // R = a\*R + b\*G + c\*B, UV/IR palette**

**-0.4 1.7 -0.3 // G = a\*R + b\*G + c\*B**

**0.08 -0.74 1.66 // B = a\*R + b\*G + c\*B**

Color correction matrix for each palette. Should be calibrated with color chart. RGB sensors normally can not produce natural colors without calibration – very useful function to see real colors of the sky. Here is the example for ASI533MC with Optolong UV/IR filter.

## **Palette: DualBand**

Next palette block.

There can be 1 or more color palette blocks.

.....

## **End Palette**

End of all color palette blocks.

**0 // enhance bright stars, 0 - disabled, 1 - star "halos"**

**20 0.02 // enhance bright stars: star "halos" radius in pixels, star "halos" strength 0..1**

Size and strength must be adjusted carefully to achieve pleasant effect. Can also be distracting.

**0.3 // highlight protection activation: 0 - disabled, 0.02..1.0 - active**

sort of HDR tonemapping for highlights, prevents saturation of bright objects. Strength can be adjusted.

**1 //reject frames with satellites, 0/1: off/on**

**0.3 //reintroduce satellites with decay, 0 for off, 0.02..0.9 faster/slower decay**

Activates detecting and rejecting of frames with satellites. Second parameter activates temporarily addition of satellite image to stack, and slow decay to no satellite after that. Makes observing more dynamic.

**0.0 //reject shaky frames, 0 for off, 1.5..3.0 factor for shaky criteria (more is robust)**

**0.0 //reject cloudy frames, 0 for off, 1.5..3.0 factor for cover criteria (more is robust)**

Experimental rejection of shaky frames (can be activated) and cloudy frames (still placeholder, has no effect).

**3.0 1.5 // zoom value for focusing/zoom mode, for video and foto mode, 1..x**

Zoom value (activated per zoom button) can be adjusted separately for "video" and "foto" mode.

**2 // focusing zoom type: 1 - simple zoom, 2 - rgb zoom, 3 - edge zoom**

Type of focusing zoom in "video" mode. 1 is simple zoom of central part, 2 is zoom of central part shown separately for RGB channels, 3 is zoom of central part and edge parts of the image.

**1.0 // zoom value for display, if needed, 1..x**

Default zoom value can be changed, if only part of the image should be shown.

Can be also used to zoom a small image circle to full height.

Can be also used to hide artefacts at borders of the stacked image (use zoom 1.03-1.05).

**x m + - p s z b // control keys for eXit, Mode, stretch +, stretch -, Palette, Save, Focusing, Background correction**

Here the control keys can be changed, if needed.

**1 // main screen activation: 0 - deactivated (eyepiece only mode), 1 - active**

Here the main screen can be deactivated – in this case eyepiece screen is automatically activated, for eyepiece-only mode. Control in this mode is only possible per keyboard.

**1 // screen GUI: 0 - none, 1 - active**

GUI with on-screen buttons can be activated here. Useful also, if no keyboard is used, e.g. for tablet PC with touch screen.

**1 // show clock waiting symbol**

Activates clock symbol on display, if SW is busy with processing after pressing a button.

**1 // show status field**

Activates status field in the bottom of main window.

**//----- AI noise reduction**

**2 // AI noise reduction activation, 0 - none, 1 - active, 0.1-0.9 - mix of filtered/unfiltered, >1 - auto**

**0.01 0.04 0.1 0.95 // automatic AI noise reduction parameters, noise min/max, strength min/max**

**0 // activate AI noise reduction after n-th stacked frame**

**train\_04.json // AI noise reduction model file**

**0 // number of CPU threads used for AI filter, 0 - automatic detection, 1..n - use parameter**

AI noise reduction activation and parameters. Active only in stacking mode, and applied only after n-th stacked frame. First parameter can also be used to define the amount of filtering, if value <1.0. First parameter can also be used to activate automatic mode (=2), where the image noise is analysed and strength of the filter is adjusted automatically. In automatic mode the strength of filter is calculated linearly between parameters strength min and max, with limits according to noise min and max parameters. Outside the limits the strength is staying at defined limits. The calculated amount of noise and filter strength is printed in the console window for convenience of tuning. Number of CPU threads is currently not used.

## //----- Eyepiece display

**0 //Eyepiece display activation, 0 for disabled, 1 for single picture, 2 for stereo**

Activation of image on second screen in full-screen mode, meant for digital eyepieces. 1 is for single picture in the middle, 2 is for stereo picture for binocular eyepiece.

**2560 1440 //Eyepiece display resolution in pixel, bigger dimension, smaller dimension**

Physical resolution of eyepiece display.

**121 68 //Eyepiece display dimensions in mm, bigger dimension, smaller dimension**

Physical dimensions of eyepiece display in millimetres.

**61 //interpupillary distance in mm**

Interpupillary distance of observer in millimetres, needed for correct placing of stereo images on display.

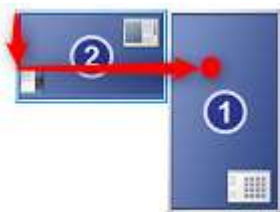
**3 //image rotation for eyepiece display: 0 (same, as main screen), 1, 2, 3 (0, 90°, 180°, 270° clockwise)**

Rotation of the whole eyepiece image, e.g. for displays with vertical orientation.

**2500 500 //coordinates of a point on second display to move image window (opencv specific)**

This is a dirty hack, because opencv has no mechanism to work with second display. These are the coordinates of a point, that should be on the second display to move the image window there. The eyepiece window first moved to these coordinates, landing on second display, and then goes full-screen. Shown configuration is good for Full-HD main screen, and second screen placed to the right of the main screen.

main      eyepiece



**0 //circular mask for eyepiece display (optic like): 0 - none, 1 - active for both video and foto mode**

Here the black circular mask can be activated, making picture more optic-device-like. Not needed, if physical round field stop is used in the eyepiece display.

**//----- Night Vision Mode**

**0 //activation of high-fps night vision mode, 0 - off, 1 - active for video mode**

Activation of Night Vision mode. If 0 – no further parameters in config file.

**CHECK //check consistency of config file**

Predefined string to check consistency of config file.

If Night Vision mode is activated, following config parameters are used.

**//----- Night vision mode**

**1 //activation of high-fps night vision mode, 0 - off, 1 - active for video mode**

Activation of Night Vision mode – high fps (e.g. tested with 10fps) in “video” mode with special averaging and noise reduction. Resembles a night vision device.

**2 // 0 - 10 frames average, 1 - Kalman smoothing, 2 - motion detection**

Type of frames averaging. 0 is averaging of last 10 frames, not recommended. 1 is experimental Kalman filtering, not recommended. 2 is motion detection, where during motion single frames are used, without motion – naturally looking exponential averaging – recommended.

**1.0 0.15 // Kalman responsiveness, alfa: for motion, beta: no motion**

Parameters for experimental Kalman filtering, not recommended.

**0.7 0.2 // motion detection: ratio threshold, mask threshold**

Parameters of motion detection. Ratio threshold – how far features in image are changed for motion detection. Mask threshold – simple threshold for feature detection.

**2 // number of frames to detect motion: 1..5**

Number of frames with changes to detect motion. More is slower but more robust.

**train\_13\_mono.onnx // AI noise reduction model file for NV mode**

Noise reduction model.

**0.0 0.8 // AI noise reduction factor for NV mode for: motion, no motion**

Activating of AI noise reduction model separately for motion and no-motion states.

## **0.5 0.0 // bilateral noise reduction factor for NV mode for: motion, no motion**

Activating of simple bilateral noise reduction separately for motion and no-motion states.

## **0.5 // gain reduction in motion**

Makes image during motion darker – looks more natural.

## **Version history**

// V0.100 beta

// initial

// V0.200 beta

// - added documentation of used libraries and licenses

// - performance upgrade

// - added information file to saved pictures

// - added color correction matrix support for rgb mode

// - added highlight protection option (similar to HDR tonemapping for highlights)

// - added optional screen GUI buttons

// - add AI noise reduction

// - optimized banding filter

// V0.3

// - added alternative zoom function to focus zoom button

// - added automatic saving of pictures

// - added black screen mode, activated with mouse click on image

// - added control of highlights part of stretching curve - star protection factor

// - added control of mixing original and AI noise filtered image

// - added correct DPI awareness mode

// V0.4

// - added separate threads for camera frame acquisition and pre-processing

// - added hot pixel detection without dark frame

// - corrected bug, working with some image sizes

// - updated ASI SDK

// V0.5

// - added support of SVBony cameras

// - added support of different bayer patterns

// - changed NN filter algorithm to prevent grid artefacts

// - changed interaction of NN filter and GUI - more responsive

// - trained new NN filters

// - added automatic mode for NN filter strength

// - updated frugally deep library

// - updated white balance and color correction

// - updated stretch tone curve

// - added parameter to highlight protection

// - released eyepiece-only mode

// - added optional clock waiting symbol

// - updated registration/alignment area for special cases

// - added midtone contrast enhancement algorithm

// - added sigma parameter for hotpixel correction

// - added black point offset parameter

// V0.7

// - added simple check for config file

// - added delete and show satellite trails

// - added support of ToupTek cameras

// - added experimental mono high-fps night vision mode

// - added finding camera per name (for multiple connected cameras)

// - corrected black level search areas

// - added output of sensor temperature

// - added hotkeys for external controller

// - added spline flat correction (for special cases)

// - added 2x ROI zoom mode

// - added optional scaling of internal image (like additional fractional bin)

// - added optional blackpoint correction for monitor and eyepiece

// - added recovery of saturated stars

// - added general configuration of color processing (white balance and color matrix)

// - added chroma preserving stretch

// - added rejecting of frames with vibrations (experimental)

// - added new focusing zoom mode (center/edges)

// - added status field for main GUI window